

Notes: Brandt, Van Biesebroeck, Wang, and Zhang (AER, 2017)

WTO Accession and Performance of Chinese Manufacturing Firms

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1 Big picture: question, contribution, and why the paper matters

Core question. How did China’s trade liberalization (especially WTO accession) affect *firm-level* and *industry-level* performance in manufacturing? The paper focuses on two performance objects that are central in empirical IO and productivity analysis:

1. **Market power:** firm-level price–cost markups (“pro-competitive” effects),
2. **Efficiency:** firm-level productivity (TFP), and how *entry/exit and reallocation* shape industry-level outcomes.

Main contribution. The paper provides a unified empirical framework where *output tariff cuts* and *input tariff cuts* have conceptually different implications:

- **Output tariff cuts** increase import competition in the *final-good market*. The empirical prediction is a decline in markups (a pro-competitive effect) and an increase in productivity (selection + within-firm improvements).
- **Input tariff cuts** reduce costs / improve access to intermediates (cost/technology channel). If cost reductions are not fully passed through to prices, markups can rise even as productivity rises.

Empirical IO angle. The paper is an empirical IO–style analysis because it measures markups and interprets tariff shocks as changes in competitive pressure and cost structure. It is also a productivity/misallocation paper because it decomposes industry-level performance changes into within-firm, between-firm reallocation, and entry/exit contributions.

2 Institutional background and key treatment variables

2.1 Two tariff objects: output vs input tariffs

The paper builds two industry-level tariff series:

- **Output tariff** $\tau_{s,t}^O$: tariffs on imported final goods in industry s (mapped from HS 8-digit products to 4-digit CIC industries).
- **Input tariff** $\tau_{s,t}^I$: a weighted average of tariffs on imported intermediate inputs used by industry s , constructed using the (Chinese) input–output table.

2.2 Identification and endogeneity concerns

The paper emphasizes two threats:

1. **Reverse causality:** productive industries might receive larger tariff cuts if policy anticipates competitiveness.
2. **Omitted variables / misattribution:** concurrent reforms may correlate with tariff changes.

The key instrument is the **maximum allowable tariff under the WTO agreement** (“bound” tariff). From 2001 onward, this agreed upper bound is used as an instrument for actual applied tariffs, exploiting that these bound rates were negotiated in advance and largely fixed by 1999.

3 Baseline empirical model for performance

3.1 Firm-level reduced-form specification (Eq. (1))

For a firm i in industry s at time t , the paper estimates:

$$y_{it} = \alpha_O \tau_{s,t-1}^O + \alpha_I \tau_{s,t-1}^I + Z_{s,t}\gamma + \gamma_t + \gamma_i + \gamma_s + \varepsilon_{it}, \quad (1)$$

where:

- y_{it} is a firm performance outcome (e.g., log markup or log TFP),
- $\tau_{s,t-1}^O$ and $\tau_{s,t-1}^I$ are *lagged* output and input tariffs (tariff rates are treated as fractions, e.g. 0.10 for 10%),
- $Z_{s,t}$ are time-varying industry controls (the paper includes controls such as sectoral output and export variables, and foreign firm share),
- γ_t are year fixed effects,
- γ_i are firm fixed effects,
- γ_s are industry fixed effects for the subset of firms that switch industries (otherwise absorbed by γ_i).

Interpretation.

- $\alpha_O > 0$ in a markup regression means *higher* output tariffs are associated with *higher* markups; therefore, *cutting* output tariffs reduces markups (pro-competitive effect).
- $\alpha_I < 0$ in a markup regression means *higher* input tariffs are associated with *lower* markups; therefore, *cutting* input tariffs raises markups (a cost channel with incomplete pass-through).
- In a productivity regression, $\alpha_O < 0$ or $\alpha_I < 0$ means tariff cuts raise productivity.

4 Measuring markups (Eq. (2))

4.1 Markup definition

The markup is defined as the price-cost wedge:

$$\mu_{it} \equiv \frac{p_{it}^Q}{MC_{it}},$$

where p_{it}^Q is the firm-specific output price and MC_{it} is marginal cost.

4.2 De Loecker–Warzynski markup formula

Using a variable input (materials/intermediates) denoted M , the paper applies:

$$\mu_{it} = \frac{p_{it}^Q}{MC_{it}} = \frac{\beta_m}{s_{it}^M}, \quad (2)$$

where β_m is the *output elasticity* of materials in the production function, and s_{it}^M is the (appropriately measured) materials expenditure share in revenue:

$$s_{it}^M = \frac{\exp(m_{it}^n)}{\exp(r_{it} - \hat{\epsilon}_{it})}. \quad (3)$$

Here m_{it}^n is (log) nominal materials expenditure, r_{it} is (log) nominal revenue, and $\hat{\epsilon}_{it}$ is the estimated transitory error term from production-function estimation.

Why this is IO-relevant. Equation (2) turns observed *cost shares* into a direct measure of *market power*, once one has an estimate of the output elasticity β_m . That is a key bridge between productivity estimation and empirical IO questions about competition.

5 Measuring productivity with imperfect price deflators (Eqs. (3)–(4))

5.1 Physical production function

A baseline (gross output) Cobb–Douglas production function in physical quantities is:

$$q_{it} = \beta_k k_{it} + \beta_l l_{it} + \beta_m m_{it} + \omega_{it} + \epsilon_{it}, \quad (4)$$

where q_{it} is log physical output, k_{it} log capital, l_{it} log labor, m_{it} log materials, ω_{it} is firm productivity, and ϵ_{it} is transitory noise.

5.2 Observed data: deflated revenue and the “price bias” problem

In many firm datasets, one observes *revenue* (nominal sales) rather than physical quantities and firm-specific prices. Let $\tilde{r}_{it}$ denote log *deflated* revenue using an industry price deflator $\bar{p}_t^Q$, and similarly $\tilde{m}_{it}$ denote deflated materials expenditure using an industry input deflator $\bar{p}_t^M$.

The paper shows that the production function written in terms of these deflated nominal variables becomes:

$$\tilde{r}_{it} = \beta_k \tilde{k}_{it} + \beta_l l_{it} + \beta_m \tilde{m}_{it} + \left(p_{it}^Q - \bar{p}_t^Q\right) - \beta_m \left(p_{it}^M - \bar{p}_t^M\right) + \omega_{it} + \epsilon_{it}. \quad (5)$$

Key insight. The residual from a deflated-revenue production function includes not only true productivity ω_{it} , but also *firm-specific price deviations* relative to the industry deflator:

$$\left(p_{it}^Q - \bar{p}_t^Q\right) - \beta_m \left(p_{it}^M - \bar{p}_t^M\right).$$

This is the “TFPR vs TFPQ” issue: revenue-based productivity mixes efficiency and price components.

5.3 Implication for growth rates (Eq. (4))

Taking differences yields:

$$\Delta TFP_{it}^R = \Delta\omega_{it} + \Delta\epsilon_{it} + \left(\Delta p_{it}^Q - \Delta\bar{p}_t^Q\right) - \beta_m \left(\Delta p_{it}^M - \Delta\bar{p}_t^M\right). \quad (6)$$

How the paper uses this. Because tariffs are mapped at a fine industry level and the same deflators are industry-level objects, the tariff variation is argued to be orthogonal to within-industry firm price deviations in expectation. Thus, tariff coefficients in productivity regressions can still be interpreted as reflecting genuine efficiency changes (plus industry-level price effects controlled by fixed effects), rather than being entirely driven by price bias.

6 Connecting productivity and markups (Eq. (5))

The paper emphasizes that changes in markups can be written as:

$$\Delta \ln \mu_{it} = \Delta p_{it}^Q - \Delta MC(\Delta\omega_{it}, \Delta p_{it}^W, \Delta q_{it}), \quad (7)$$

where p_{it}^W denotes factor prices (wages, rental rates, etc.).

Cobb–Douglas intuition (special case). Under constant returns and a Cobb–Douglas technology with all inputs variable, a common approximation is:

$$\Delta \ln \mu_{it} \approx \Delta\omega_{it} + \Delta p_{it}^Q - \beta^W \Delta p_{it}^W,$$

so markups rise when productivity increases (lower marginal cost) unless output prices fall one-for-one.

7 Industry-level decomposition with entry and exit (Eq. (6))

7.1 Why a decomposition?

The firm-level regressions show how an average firm changes. But aggregate performance can improve even if incumbents do not, due to:

1. reallocation toward better firms among incumbents,

2. entry of high-performing firms,
3. exit of low-performing firms.

7.2 Griliches–Regev / Haltiwanger-style decomposition

Let y_{it} be a firm outcome (e.g., $\log \mu$ or $\log TFP^R$). Let $y_{s,t}$ be the output-share weighted industry mean in period t . Over a multi-year interval from $t - k$ to t , the change is decomposed as:

$$\Delta y_{s,t} = \sum_{i \in C} \left(\bar{s}_i \Delta y_{it} + \Delta s_{it} [\bar{y}_i - y_{s,t-k}] \right) + \sum_{i \in E} s_{it} [y_{it} - y_{s,t-k}] - \sum_{i \in X} s_{i,t-k} [y_{i,t-k} - y_{s,t-k}], \quad (8)$$

where:

- C = continuing firms, E = entrants, X = exiters,
- s_{it} is firm i 's output share in industry s at time t ,
- $\bar{s}_i = (s_{it} + s_{i,t-k})/2$ is the time-average share,
- $\bar{y}_i = (y_{it} + y_{i,t-k})/2$ is the time-average performance.

Economic meaning of the terms.

- **Within:** $\sum_{i \in C} \bar{s}_i \Delta y_{it}$ — improvements among continuers.
- **Between (reallocation):** $\sum_{i \in C} \Delta s_{it} [\bar{y}_i - y_{s,t-k}]$ — market share shifts toward higher- y firms.
- **Entry:** $\sum_{i \in E} s_{it} [y_{it} - y_{s,t-k}]$ — whether entrants outperform the initial industry mean.
- **Exit:** $-\sum_{i \in X} s_{i,t-k} [y_{i,t-k} - y_{s,t-k}]$ — whether exiting firms were below the initial mean (a positive contribution if low performers exit).

8 Figures: what they show and how to read them

8.1 Figure 1: tariffs on outputs and inputs fall and compress over time

Figure 1 plots the median output tariff and median input tariff across industries, plus interquartile ranges. Two facts stand out:

1. Both output and input tariffs decline strongly from the mid-1990s to 2007.
2. Dispersion compresses: high-tariff industries experience larger cuts (tariff convergence).

This supports the idea that much of the variation is “predetermined” by initial protection levels.

8.2 Figure 2: other protection measures decline too

Figure 2 reports the fraction of industries with (i) high tariffs, (ii) any FDI restriction, and (iii) any non-tariff barrier. All three decline, with a notable downward shift around WTO accession. The paper therefore uses tariffs as a clean, detailed proxy for liberalization intensity.

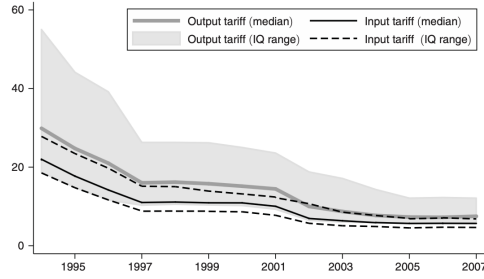


FIGURE 1. EVOLUTION OF IMPORT TARIFFS ON EACH SECTOR'S OUTPUT AND INPUTS

Notes: Product-level (output) tariffs at the HS 8-digit level are mapped into 4-digit CIC industries. Multiplying the tariff vector with the 2002 Input-Output table generates input tariffs approximately at the 3-digit CIC level. The solid lines show the median tariffs across industries, the shaded area (for output tariffs) and dashed lines (for input tariffs) show the interquartile range.

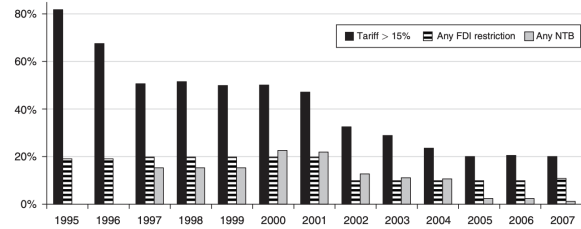


FIGURE 2. FRACTION OF MANUFACTURING SECTORS SUBJECT TO HIGH TARIFFS, NON-TARIFF BARRIERS, OR INVESTMENT RESTRICTIONS

Notes: The annual statistics average sectoral dummies across 424 4-digit (CIC) manufacturing sectors. Tariff rates are from WITS; FDI restrictions and NTB barriers are from annual circulars of the Ministry of Foreign Trade and Economic Cooperation and the Ministry of Commerce.

8.3 Figure 3: tariff changes are strongly predicted by initial tariff levels

Figure 3 plots tariff changes against initial tariffs (1992–2007 and 2001–2007). The strong negative slope implies that industries with higher initial protection experienced larger cuts. This is crucial for instrument relevance: bound (maximum) tariffs are closely linked to initial tariffs and thus to subsequent applied tariffs.

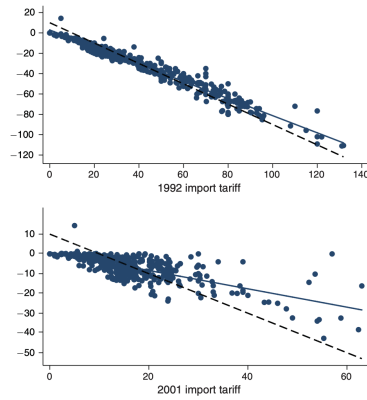


FIGURE 3. CHANGES IN IMPORT TARIFFS BY INITIAL TARIFF LEVEL

Notes: Observations are 4-digit CIC manufacturing sectors. The solid line is the regression line of tariff changes on initial tariffs (with a slope of -0.84 over 1992–2007 and -0.46 over 2001–2007); the dashed line has a slope of -1 (for comparison).

TABLE 1—DETERMINANTS OF CHANGES IN TARIFF PROTECTION

	Change in tariffs from 1998 to 2007 ($\tau_{2007} - \tau_{1998}$)				Tariff level 2002–2007
	(1)	(2)	(3)	(4)	(5)
TFP_{1998}	-0.428 (0.293)		-0.852 (0.448)	-0.467 (0.499)	
$TFP_{1998} - TFP_{1995}$		0.371 (0.302)	0.822 (0.450)	0.335 (0.532)	
$TFP_{1998} \times (TFP_{1998} - TFP_{1995})$			0.005 (0.111)	-0.019 (0.127)	
$EXP_{1998} - EXP_{1995}$				0.249 (0.822)	
$EXP_{2007} - EXP_{1998}$				0.001 (0.026)	
<i>Trade categories (BEC)</i>					
Intermediates (dummy)				0.043 (0.032)	
Capital goods (dummy)				0.017 (0.033)	
Consumer goods (dummy)				0.007 (0.031)	
<i>US industry characteristics</i>					
Differentiated goods (Rauch dummy)				-0.008 (0.012)	
Sales expenditure intensity				0.053 (0.021)	
R&D intensity				-0.497 (0.395)	
Advertising intensity				-0.315 (0.193)	
<i>Chinese industry characteristics</i>					
Top 4 market share				-0.016 (0.023)	
log employment				-0.837 (0.343)	-0.003 (0.001)
log (K/Y) ratio				0.925 (0.963)	
SOE share in sectoral sales				-0.011 (0.019)	0.004 (0.002)
Elementary education share				0.133 (0.099)	
Maximum tariff level					0.524 (0.018)
Sector (2-digit) fixed effects	Yes		Yes	Yes	Yes
Industry (4-digit) fixed effects					
R^2	0.35	0.01	0.41	0.49	0.98
Observations	410	410	410	370	2,484

Notes: Estimates in column 4 additionally contain year fixed effects and the same additional controls as used in firm-level regressions below (sectoral output, export, and the foreign firm share). Standard errors in parentheses.

9 Tables: interpretation and takeaways

9.1 Table 1: determinants of tariff changes and instrument relevance

What Table 1 does. It regresses (i) the change in tariffs from 1998 to 2007 and (ii) the tariff level in 2002–2007 on pre-liberalization industry characteristics.

How to read the main messages.

- The coefficients on pre-period productivity measures (e.g. TFP_{1998} and $TFP_{1998} - TFP_{1995}$) are not robustly significant, suggesting tariff cuts are not systematically “targeted” to industries with special productivity trends.
- The “maximum tariff level” is a powerful predictor of realized tariffs (column (5) shows a very high R^2), supporting the strength of the WTO bound-tariff instrument.

9.2 Table 2: tariffs change imports and domestic prices (sanity checks + mechanism)

Panel A (imports). The dependent variable is log ordinary imports at the product level (HS6), 2000–2007. The negative coefficient on lagged import tariffs means tariff reductions raise imports, consistent with trade theory.

Panel B (domestic prices). The dependent variable is an industry price index. Positive coefficients imply higher tariffs are associated with higher domestic prices; hence tariff cuts tend to reduce domestic prices. This supports the pro-competitive mechanism: more openness lowers the domestic price level and pressures markups.

Ordinary trade imports into China (in logs) at the product level (6-digit HS)					
	All goods (1a)	Materials (2a)	Intermediate inputs (3a)	Capital goods (4a)	Consumer goods (5a)
<i>Panel A. Imports, 2000–2007</i>					
Import tariff (lagged)	−3.112 (0.331)	−1.559 (0.826)	−4.349 (0.668)	−2.325 (0.692)	−1.980 (0.631)
Observations	127,049	8,458	73,351	19,013	26,061
Year fixed effects	Yes	Yes	Yes	Yes	Yes
Product fixed effects	Yes	Yes	Yes	Yes	Yes
<i>Notes:</i> All coefficients are estimated using separate regressions based on information of firm-level trade transactions from China Customs that are aggregated to the product level. Duty-free imports are excluded. The regressions allow for a different elasticity for imports entering through trade intermediaries (not reported). The instrument, from 2001 onward, is the maximum allowable tariff under the WTO agreement. Standard errors in parentheses are clustered at the product level.					
Index of the Chinese domestic price level at the industry level (4-digit CIC)					
	All goods (1b)	Materials (2b)	Intermediate inputs (3b)	Capital goods (4b)	Consumer goods (5b)
<i>Panel B. Domestic price level, 1998–2007</i>					
(i) Preferred series: 4-digit where available and 2-digit elsewhere					
Import tariff (lagged)	0.296 (0.090)	0.320 (0.212)	0.492 (0.191)	0.174 (0.088)	0.011 (0.106)
(ii) Alternative series (1): constructed from the firm-level information					
Import tariff (lagged)	0.283 (0.100)	−0.309 (0.212)	0.493 (0.204)	0.197 (0.091)	0.049 (0.112)
(iii) Alternative series (2): aggregate deflators constructed by NBS for 2-digit sectors					
Import tariff (lagged)	0.255 (0.092)	0.298 (0.244)	0.442 (0.169)	0.182 (0.158)	−0.085 (0.089)
Observations	4,240	70	1,950	1,180	980
Year fixed effects	Yes	Yes	Yes	Yes	Yes
Industry or sector fixed effects	Yes	Yes	Yes	Yes	Yes

Notes: All coefficients are estimated using separate regressions for 4-digit industries. The instrument, from 2001 onward, is the maximum allowable tariff under the WTO agreement. In panel (i), the price index from 1998 to 2003 is calculated by aggregating firm-level price changes to the 4-digit industry level as described in Section IV; the index is extended to the end of the sample period using a sectoral (2-digit) price index taken from China’s *Statistical Yearbook*. In panel (ii), the detailed price index for 1998 to 2003 is calculated omitting fewer outlier observations. In panel (iii), the 2-digit sector-level price index is used throughout. Standard errors in parentheses are clustered at the industry level.

	No weights		Within-industry output share weights		Output weights	
	(1a)	(2a)	(3a)	(4a)	(5a)	(6a)
<i>Panel A. Markups, 1998–2007</i>						
Output tariff (lagged)	0.045 (0.041)	0.058 (0.044)	0.07 (0.043)	0.095 (0.051)	0.375 (0.110)	0.543 (0.120)
Input tariff (lagged)	−0.227 (0.106)	−0.969 (0.074)	0.016 (0.169)	−0.733 (0.094)	−0.651 (0.354)	−1.173 (0.181)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Sector-year fixed effects		Yes		Yes		Yes
Observations	1,293,495	1,295,372	1,211,861	1,213,586	1,293,495	1,295,372
<i>Panel B. Productivity (TFP), 1998–2007</i>						
Output tariff (lagged)	−0.369 (0.139)	−0.316 (0.097)	−0.176 (0.088)	−0.107 (0.062)	−0.263 (0.155)	−0.387 (0.115)
Input tariff (lagged)	−0.806 (0.385)	−1.031 (0.167)	1.790 (0.323)	−1.621 (0.144)	−2.395 (0.616)	−0.682 (0.249)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Sector-year fixed effects		Yes		Yes		Yes
Observations	1,275,041	1,276,922	1,194,720	1,196,451	1,275,041	1,276,922

Notes: Instrument from 2001 onward is the maximum allowable tariff under the WTO agreement. The sample only includes firms that did not switch industry. Standard errors in parentheses are clustered two-ways at the industry-year and the firm level.

9.3 Table 3: main firm-level results — markups and productivity

Table 3 estimates (1) using firm-level log markups (Panel A) and log productivity (Panel B), 1998–2007. The table reports three weighting schemes: unweighted, within-industry output-share weights, and output weights.

Panel A: Markups.

- **Output tariffs:** coefficients are generally positive. This means lowering output tariffs reduces markups (pro-competitive effect). The effect is economically larger in output-weighted specifications, suggesting the average markup decline is concentrated in high-output firms.
- **Input tariffs:** coefficients are generally negative. This means lowering input tariffs raises markups, consistent with marginal-cost reductions not being fully passed through to prices.

Panel B: Productivity (TFP).

- **Output tariffs:** coefficients are negative across specifications, implying output tariff reductions increase productivity.
- **Input tariffs:** most specifications show negative coefficients (input tariff reductions raise productivity), though magnitudes can differ depending on weighting and fixed effects, reflecting the coarser variation in input tariffs (constructed at a higher aggregation level).

9.4 Table 4: industry-level decomposition — which margins drive aggregate changes?

Table 4 applies the decomposition (8) to industry-level changes in markups (Panel A) and productivity (Panel B), then regresses each decomposition term on tariff changes.

Markups (Panel A).

- **Output tariff reductions** lower markups primarily through the **within** term (continuing firms reduce markups).
- **Input tariff reductions** raise markups, with an important role for the **entry** term (industries experiencing larger input tariff cuts see entrants contributing more to markup growth).

Productivity (Panel B).

- Both **output** and **input** tariff reductions raise productivity.
- The **entry** term is quantitatively large: tariff cuts are strongly associated with entrants having productivity above the initial industry mean. This is an explicit selection/composition channel.

TABLE 4—DECOMPOSING THE EFFECT OF TARIFFS ON INDUSTRY PERFORMANCE

	Total effect (1a)	Within (2a)	Between (3a)	Entry (4a)	–Exit (5a)
<i>Panel A. Price-cost markups, 1998–2007</i>					
Change in output tariff	0.152 (0.066)	0.122 (0.043)	0.010 (0.013)	0.014 (0.039)	0.005 (0.009)
Change in input tariff	–0.292 (0.149)	–0.045 (0.098)	0.040 (0.030)	–0.282 (0.088)	–0.005 (0.021)
Industry fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	844	844	844	844	844
	(1b)	(2b)	(3b)	(4b)	(5b)
Change in output tariff	0.221 (0.049)	0.150 (0.032)	–0.023 (0.009)	0.095 (0.026)	0.001 (0.006)
Change in input tariff	–0.400 (0.127)	–0.146 (0.083)	0.068 (0.024)	–0.319 (0.068)	–0.002 (0.015)
Industry-year weights	Yes	Yes	Yes	Yes	Yes
Observations	844	844	844	844	844
	(1c)	(2c)	(3c)	(4c)	(5c)
<i>Panel B. Productivity, 1998–2007</i>					
Change in output tariff	–0.569 (0.137)	–0.200 (0.074)	0.042 (0.028)	–0.428 (0.106)	–0.017 (0.016)
Change in input tariff	–1.160 (0.312)	–0.297 (0.168)	0.246 (0.065)	–1.080 (0.241)	–0.030 (0.036)
Industry fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	844	844	844	844	844
	(1d)	(2d)	(3d)	(4d)	(5d)
Change in output tariff	–0.465 (0.141)	–0.209 (0.071)	0.049 (0.022)	–0.302 (0.097)	–0.003 (0.010)
Change in input tariff	–1.236 (0.363)	–0.551 (0.184)	0.170 (0.056)	–0.837 (0.252)	–0.019 (0.026)
Industry-year weights	Yes	Yes	Yes	Yes	Yes
Observations	844	844	844	844	844

Notes: Separate regressions in each numbered column. The instrument from 2001 onward is the change in the maximum allowable tariff under the WTO agreement. Dependent variables are the terms of a linear decomposition of industry-level change in markup or productivity. The sample pools annualized growth rates over 2 subperiods (1998–2001 and 2001–2007) that cover 424 and 420 industries, respectively. Standard errors in parentheses.

TABLE 5—USE OF IMPORTED INTERMEDIATES (Percent)

	All imports (1)	Processing trade (2)	Ordinary trade (3)
<i>Panel A. Fraction of firms using imported intermediates</i>			
2000	9.6	6.9	6.6
2007	10.9	8.7	7.1
<i>Panel B. Share of imported intermediates in total input use for the sample of matched firms</i>			
2000	29.1	19.7	9.3
2007	27.3	17.5	9.8
<i>Panel C. Share of imported intermediates in total input use for the entire manufacturing sector</i>			
2000	13.4	6.2	7.2
2007	13.5	5.9	7.5

Notes: Calculations in panels A and B are based on a sample of firms merged between the NBS firm-level data and the customs' trade-transactions data; in panel C, the numerator is calculated using all firms from the customs' data and the denominator uses the sum of the aggregate input use over all firms in the NBS data or using the estimate for manufacturing from the national accounts.

9.5 Table 5: imported intermediates — how important is the direct imported-input channel?

Table 5 provides descriptive evidence on the prevalence and intensity of imported intermediate use. The key message is that, while imported intermediates are used (especially in processing trade), the aggregate share of imported intermediates in total input use changes only modestly between 2000 and 2007. This suggests that the productivity effects of input tariff cuts may reflect not only *direct* imported-input usage but also broader cost/technology adjustments and equilibrium changes.

9.6 Table 6: competition, growth, exit, and CEO replacement (agency channel)

Table 6 studies whether increased competition affects firm outcomes and managerial discipline, distinguishing SOEs from other firms. The patterns support an **agency/effort** mechanism: competition (lower output tariffs) is associated with lower survival probabilities (tougher selection) and changes in CEO replacement dynamics, with different responses for SOEs vs non-SOEs.

9.7 Table 7: tariff effects by firm status (incumbent vs entrant vs exiting)

Table 7 interacts tariffs with firm status and asks whether liberalization changes *levels* of markups and productivity differently for incumbents, entrants, and exiters. The main qualitative message is that lower output tariffs are associated with higher productivity levels across statuses, and the entrant contribution to productivity (highlighted in Table 4) is consistent with entrants in liberalized industries being more productive relative to the initial industry mean.

TABLE 6—EFFECT OF TARIFFS ON GROWTH, EXIT, AND MANAGEMENT REPLACEMENT

	Output growth (year on year) (1)	Dummy for firm survival (2)	Dummy for CEO replacement (3)	Dummy for CEO replacement (4)
SOE dummy	-0.1187 (0.0059)	-0.0776 (0.0035)	0.0537 (0.0013)	0.0606 (0.0025)
Output tariff (lagged)	-0.0110 (0.0004)	0.0005 (0.0002)		0.0004 (0.0002)
Output tariff (lagged) $\times$ SOE	-0.0013 (0.0004)	-0.0004 (0.0002)		-0.0004 (0.0002)
Output growth			-0.0099 (0.0008)	-0.0055 (0.0014)
Output growth $\times$ SOE			-0.0057 (0.0016)	-0.0075 (0.0029)
Output growth $\times$ output tariff (lagged)				-0.0002 (0.0001)
Output growth $\times$ output tariff (lag) $\times$ SOE				0.0004 (0.0002)
Firm fixed effects, year fixed effects, control variables	Yes	Yes	Yes	Yes
Observations	1,318,033	1,341,456	924,693	841,921

Notes: All regressions use data for 1998 to 2007 and control for capital-intensity and CEO tenure. Standard errors in parentheses are clustered two-ways, at the industry-year and at the firm level.

TABLE 7—EFFECT OF TARIFFS ON THE LEVEL OF MARKUP AND PRODUCTIVITY BY FIRM STATUS

	Markup (1)	Productivity (2)
Output tariff (lagged)		
$\times$ Incumbent	0.004 (0.040)	-0.454 (0.143)
$\times$ Entrant	0.017 (0.054)	-0.568 (0.202)
$\times$ Exiting firm	-0.010 (0.044)	-0.373 (0.148)
Input tariff (lagged)		
$\times$ Incumbent	-0.185 (0.107)	-0.284 (0.386)
$\times$ Entrant	-0.153 (0.164)	-0.914 (0.507)
$\times$ Exiting	-0.083 (0.141)	-0.363 (0.403)
Industry fixed effects	Yes	Yes
Year by firm-status interaction fixed effects	Yes	Yes
Observations	901,048	904,960

Notes: Sample covers the period 1999–2006 and only includes firms that do not switch industry and survive for at least two years. For entrants and exiting firms we exclude, respectively, their first and last year of operation to avoid calculating productivity using output for less than a full year. Instrument from 2001 onward is the maximum allowable tariff under the WTO agreement. Standard errors in parentheses are clustered two ways at the industry-year and at the firm level.

10 Mechanism summary

Step 1: output tariff cuts $\Rightarrow$ tougher product-market competition. Tariff cuts increase import penetration and reduce domestic prices (Table 2 and Figures 1–2). This reduces markups (Table 3) and tightens the survival constraint (Table 6), accelerating selection.

Step 2: selection and reallocation amplify productivity. Industry-level decomposition shows large entry contributions to productivity (Table 4), consistent with more productive entrants or a higher productivity threshold for entry (Table 7).

Step 3: input tariff cuts $\Rightarrow$ cost reductions and technology access. Lower input tariffs reduce marginal costs; if pass-through is incomplete, markups can rise (Table 3). Productivity rises as well (Tables 3–4), through channels that may include improved inputs, quality upgrading, and complementary domestic reforms.

Step 4: interaction with domestic institutions. SOE behavior and CEO replacement evidence (Table 6) suggests that competition interacts with governance/agency frictions, shaping within-firm improvements.

11 Replication

1. **Construct tariffs:** map HS8 output tariffs to 4-digit CIC industries; build input tariffs using IO weights.
2. **Estimate production functions:** obtain β_m for each sector and recover $\hat{\epsilon}_{it}$.
3. **Compute markups:** apply (2)–(3).
4. **Compute revenue productivity:** residual from (5); use growth logic (6) for interpretation.
5. **Firm-level regressions:** estimate (1) with firm FE + year FE; instrument tariffs post-2001 with bound tariffs.

6. **Industry decomposition:** build within/between/entry/exit terms (8); regress each on tariff changes (Table 4).
7. **Mechanism checks:** imports and prices regressions (Table 2), imported intermediates (Table 5), agency/CEO replacement (Table 6), entrant/exit status heterogeneity (Table 7).

12 Conclusion

12.1 Summary

This paper’s core conclusion is that China’s WTO-driven trade liberalization changed firm performance through *two distinct tariff channels* that have *different implications for markups and productivity*:

1. **Output-tariff cuts (final-good import competition):** lower *markups* but raise *productivity*. The pro-competitive effect dominates: import competition compresses price–cost margins for incumbents, while tougher selection and improved incentives raise efficiency.
2. **Input-tariff cuts (intermediate-input cost channel):** raise *markups* and raise *productivity*. Cheaper inputs lower marginal cost; if pass-through is incomplete, markups increase. At the same time, productivity rises, but not primarily because firms suddenly import vastly more intermediates; rather, the paper argues that lower input tariffs free up resources and support productivity-enhancing investments.

A methodological conclusion is equally central: using detailed price deflators allows the authors to separately identify average effects on *markups* and *productivity* (which are otherwise confounded when output is measured only by revenue).

12.2 Economic intuition: why output and input tariffs move markups differently

(i) Output tariffs: competition squeezes rents but can raise efficiency. A reduction in output tariffs increases foreign competitive pressure in the product market. This pushes *prices* down relative to marginal cost, reducing markups. At the same time, lower markups and heightened survival risk induce firms to improve operations. In the Chinese context, an important mechanism is that competition mitigates agency problems (especially in SOEs), raising within-firm productivity among incumbents. In addition, sectors with larger liberalization experience *more positive selection of entrants*, so entry raises industry productivity even if incumbent productivity improvements are modest.

(ii) Input tariffs: cost relief can increase markups when pass-through is incomplete. Input-tariff cuts reduce marginal costs. If product-market prices do not fall one-for-one (incomplete pass-through), the price–cost wedge increases, and markups rise. The key point is that a markup increase here does *not* contradict pro-competitive logic; it reflects a cost shock translating partly

into higher margins rather than fully into lower prices. Those higher margins can then relax internal constraints and finance upgrading, contributing to productivity growth.

12.3 Quantitative anchors from the main tables (how to quote magnitudes cleanly)

Firm-level effects (Table 3). In the output-weighted specification with sector-year fixed effects, the estimated coefficients are:

$$\hat{\alpha}_O^\mu = 0.543, \quad \hat{\alpha}_I^\mu = -1.173 \quad (\text{Panel A: log markups}),$$

$$\hat{\alpha}_O^{TFP} = -0.387, \quad \hat{\alpha}_I^{TFP} = -0.682 \quad (\text{Panel B: log productivity}).$$

A useful back-of-the-envelope mapping is that a tariff change $\Delta\tau$ implies an approximate percent change $100 \cdot \hat{\alpha}\Delta\tau$ in the outcome (since the dependent variable is in logs). For example, a 10 p.p. cut in output tariffs ($\Delta\tau_O = -0.10$) implies:

$$\begin{aligned} \Delta \log \mu &\approx 0.543(-0.10) = -0.054 \Rightarrow \text{markups fall by about 5\%,} \\ \Delta \log TFP &\approx (-0.387)(-0.10) = 0.039 \Rightarrow \text{TFP rises by about 4\%.} \end{aligned}$$

A 10 p.p. cut in input tariffs ($\Delta\tau_I = -0.10$) implies:

$$\begin{aligned} \Delta \log \mu &\approx (-1.173)(-0.10) = 0.117 \Rightarrow \text{markups rise by about 12\%,} \\ \Delta \log TFP &\approx (-0.682)(-0.10) = 0.068 \Rightarrow \text{TFP rises by about 7\%.} \end{aligned}$$

(These are illustrative magnitudes; the paper emphasizes that input-tariff variation is constructed at coarser industry levels, so specifications with sector-year fixed effects are especially informative.)

Industry decomposition: which margins matter (Table 4). The decomposition shows a sharp separation across margins:

- **Markups:** the output-tariff effect is mostly *within-firm* (incumbents), consistent with pro-competitive pressure. For instance, in Panel A with industry fixed effects, the total output-tariff coefficient is 0.152 and the within component is 0.122.
- **Productivity:** the tariff effect is largely an *entry* channel. In Panel B with industry fixed effects, the total output-tariff coefficient is -0.569 and the entry component is -0.428 ; for input tariffs, the total is -1.160 and the entry component is -1.080 . Hence, entry contributes more than half (often much more) of the total productivity effect.

Why the input-variety mechanism is limited in China (Table 5). A key China-specific empirical fact is that the overall share of imported intermediates in total input use is essentially flat: for the entire manufacturing sector it is 13.4% in 2000 and 13.5% in 2007. Even ordinary-import intermediates rise only modestly (from 7.2% to 7.5%). This supports the paper’s interpretation

that input-tariff cuts affect productivity mainly through margins other than a dramatic expansion of imported input use.

12.4 Broader implications and a clear limitation

Implications. The results highlight a complementary role of **trade liberalization** and **domestic reforms that ease entry and reshape firm ownership**. They also speak to welfare: channels such as pro-competitive markup compression, within-firm productivity improvements, and entrant selection are not fully captured by “sufficient-statistic” approaches that abstract from variable markups and extensive-margin reallocation.

Limitation (as stated in the paper). A central limitation is the lack of firm-specific prices; even with detailed deflators, production-function estimation requires assumptions. As a consequence, the analysis identifies *average* industry-level effects rather than fully flexible heterogeneous treatment effects.

12.5 Takeaway

China’s WTO liberalization raised manufacturing productivity through within-firm improvements and especially positive entrant selection, while its impact on markups depends on whether liberalization increases product-market competition (output tariffs $\downarrow \Rightarrow$ markups $\downarrow$) or lowers input costs with imperfect pass-through (input tariffs $\downarrow \Rightarrow$ markups $\uparrow$).